1.1.6 Getting Used to *Mathematica*

This section has given you a first glimpse of *Mathematica*. If you are used to other computer systems, you will be beginning to see some of the ways that *Mathematica* is different.

Arguments of functions are given in *square brackets*.
Names of built-in functions have their first letters capitalized.
Multiplication can be represented by a space.
Powers are denoted by `^`.
Numbers in scientific notation are entered for example as `2.5 10^-4`.

Some superficial differences between *Mathematica* and other systems.

At first, you may find some of these differences confusing. You should realize, however, that there are good reasons for *Mathematica* to be set up the way it is.

Many systems use parentheses both for grouping, and for giving function arguments. In *Mathematica*, parentheses are used strictly for grouping, and square brackets are used for function arguments. The standard mathematical use of parentheses for functions is often confusing. For example, does $c(1+x)$ mean `c[1+x]` or `c*(1+x)`? Many systems insist that multiplication be indicated by an explicit star. The use of square brackets for function arguments in *Mathematica* makes this unnecessary, and allows you to type expressions like `c(1+x)` in standard mathematical notation.

You may also wonder about the use of long names for built-in *Mathematica* functions. Would it not be easier if the function to generate a pseudorandom number were called `Rand`, rather than `Random`?

There is a convention in *Mathematica* that all function names are spelled out as full English words, unless there is a standard mathematical abbreviation for them. The great advantage of this scheme is that it is *predictable*. Once you know what a function does, you will usually be able to guess exactly what its name is. If the names were abbreviated, you would always have to remember exactly which shortening of the standard English words was used.

Most implementations of *Mathematica* are capable of *command completion* (see Section 1.3). Once you have typed the beginning of a function name, you can get the system to try and complete it. As a result, you very rarely end up having to

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type the whole function name.

Another feature of built-in *Mathematica* names is that they all start with capital letters. In later sections, you will see how to define variables and functions of your own. The capital letter convention makes it easy to distinguish built-in objects. If *Mathematica* used `i` to represent $\sqrt{-1}$, then you would never be able to use `i` as the name of one of your variables.

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